Question 6.7 on Quasi-Local Operators Was Answered by Ozawa

Literature-resolution packet for arXiv:1809.00532

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Status: literature already answered. Ozawa proves that every uniformly locally finite metric space containing a sequence of expanders has a quasi-local operator on $\ell^2(X)$ that is not approximable by finite-propagation operators. This is exactly the example requested in Question 6.7 of Špakula–Zhang.

1 The source question

Špakula and Zhang [1] prove equality of the uniform Roe and quasi-local algebras under Property A and ask on PDF page 20:

> **Question 6.7.** Is there an example of a space X with bounded geometry and a quasi-local operator on (say) $\mathcal{B}(\ell^2(X))$ which does not belong to the uniform Roe algebra $C^*_u(X)$?

Here $C^*_u(X)$ is the norm closure of the finite-propagation operators, while $C^*_{ql}(X)$ is the algebra of quasi-local operators. Thus the question asks whether $C^*_u(X) \subsetneq C^*_{ql}(X)$ can occur.

Question 6.6. Is it possible to generalise the characterisation of Property A (Proposition 6.3) to any $p \in [1, \infty)$?

Second, in the case of $p \in (1, \infty)$ or especially when $p = 2$, is the assumption of Property A really necessary for all the quasi-local operators to be approximable by operators with finite propagation? In other words:

Question 6.7. Is there an example of a space X with bounded geometry and a quasi-local operator on (say) $\mathfrak{B}(\ell^2(X))$ which does not belong to the uniform Roe algebra $B_u^2(X)$ (usually denoted $C_u^*(X)$ in the literature)?

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2 Ozawa's exact answer

Ozawa's paper [2] studies the same two algebras on uniformly locally finite (equivalently, bounded-geometry in this setting) metric spaces. Its abstract says that the results yield a quasi-local operator not approximable by finite-propagation operators. The introduction proves:

> **Theorem A.** The C^*-algebra $\prod_n M_n$ does not embed into the uniform Roe algebra $C_u^*[X]$ of any uniformly locally finite metric space X.
>
> **Theorem B.** The C^*-algebra $\prod_n M_n$ embeds into the quasi-local algebra $C_{ql}^*[X]$ of a uniformly locally finite metric space X, provided that X contains a sequence of expanders.
>
> **Corollary C.** For every uniformly locally finite metric space X containing a sequence of expanders, the inclusion $C_u^*[X] \subset C_{ql}^*[X]$ is proper. Equivalently, there is a quasi-local operator which is not approximable by finite-propagation operators.

Choosing any bounded-geometry coarse disjoint union of finite expanders gives the requested X. Properness produces an element $T \in C_{ql}^*(X) \setminus C_u^*(X)$, which is precisely the requested operator. There is no mismatch in category or closure: both papers work on $\ell^2(X)$ and define the uniform Roe algebra as the norm closure of finite-propagation operators.

The connection is explicit rather than inferred only from terminology. Ozawa cites Špakula–Zhang as [SZ], states that they prove equality for Property A spaces, and cites page 20 of the earlier literature when framing the equality problem.

see also [BB+, En, KL+, LN+, ŠT, ŠZ] and references therein). It is proved in [ŠZ] that a large class of ulf metric spaces, namely those with property A, satisfy the equality $\mathrm{C}^*_{\mathrm{u}}[X] = \mathrm{C}^*_{\mathrm{ql}}[X]$. See Section 4 for the definition of property A and an alternative proof of this fact. In this paper, we prove that the inclusion $\mathrm{C}^*_{\mathrm{u}}[X] \subset \mathrm{C}^*_{\mathrm{ql}}[X]$ can be proper in general. The proof takes a roundabout way and goes by studying the embeddability of the C^*-algebra $\prod_n \mathbb{M}_n$ of the ℓ_∞-direct sum of matrix algebras. Whether embeddings are unital or not will make no essential difference.

Theorem A. *The C^*-algebra $\prod_n \mathbb{M}_n$ does not embed into the uniform Roe algebra $\mathrm{C}^*_{\mathrm{u}}[X]$ of any ulf metric space X.*

Theorem B. *The C^*-algebra $\prod_n \mathbb{M}_n$ embeds into the quasi-local algebra $\mathrm{C}^*_{\mathrm{ql}}[X]$ of a ulf metric space X, provided that X contains a sequence of expanders.*

See Section 3 for the definition of expanders. The above results answer the problem posed in [BB+], where it is proved that non-atomic von Neumann algebras do not embed into quasi-local algebras, leaving the possibility for the atomic von Neumann algebra $\prod_n \mathbb{M}_n$ open.

Corollary C. *For any ulf metric space X that contains a sequence of expanders, the inclusion $\mathrm{C}^*_{\mathrm{u}}[X] \subset \mathrm{C}^*_{\mathrm{ql}}[X]$ is proper. In other words, there exists a quasi-local operator that is not approximable by finite propagation operators.*

We remark that this corollary holds as well in the "non-uniform" setting (see Chapter 3 in [Ro2] for the definition and, for a given Hilbert X-module H, consider an X-embedding $\ell_2 X \subset H$). Recall that property A is a kind of amenability condition and a sequence of expanders is the most prominent obstruction to it (see e.g., Sections 4 & 5 in [NY]). It seems natural to expect that $\prod_n \mathbb{M}_n$ embeds into the quasi-local algebra and hence $\mathrm{C}^*_{\mathrm{u}}[X] \neq \mathrm{C}^*_{\mathrm{ql}}[X]$ as soon as X does not have property A.

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2. Proof of Theorem A

The proof of Theorem A is motivated by an operator space theoretic perspective that the matrix algebras are hard to embed completely isomorphically into commutative C^*-algebras.

For every Banach space E, we denote by $(E)_1$ the closed unit ball of E. For every projection p, we put $p^\perp := 1 - p$.

3 Audit and scope

The theorem match is exact:

- bounded geometry corresponds to Ozawa's uniformly locally finite hypothesis;
- containing a sequence of expanders supplies explicit eligible spaces;
- proper inclusion supplies a quasi-local operator outside the uniform Roe algebra;
- being outside $C_u^*(X)$ means not being norm-approximable by finite-propagation operators.

Therefore Question 6.7 has a complete affirmative literature answer. No new proof or counterexample should be promoted for that question. Ozawa's result does not answer the distinct p-generalization asked in Question 6.6.

Search audit. The run's cheap indexes contained no prior packet for arXiv:1809.00532. A bounded exact-title and keyword search for quasi-local operators outside uniform Roe algebras located arXiv:2310.03677. Its abstract, Theorems A–B, Corollary C, and citations to Špakula–Zhang were checked against the official arXiv PDF.

Human review. Confirm the terminology identification "uniformly locally finite" with the source's bounded-geometry convention and the literal statement of Corollary C. The substantive match is otherwise direct.

Answer-paper abstract page

EMBEDDINGS OF MATRIX ALGEBRAS INTO UNIFORM ROE ALGEBRAS AND QUASI-LOCAL ALGEBRAS

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Abstract. We answer the recent problem posed by Baudier, Braga, Farah, Vignati, and Willett that asks whether the ℓ_∞-direct sum of the matrix algebras embeds into the uniform Roe algebra or the quasi-local algebra of a uniformly locally finite metric space. The answers are no and yes, respectively. This yields the existence of a quasi-local operator that is not approximable by finite propagation operators.

1. Introduction

Throughout this paper, we are interested in a (discrete) metric space X that is *uniformly locally finite*, or *ulf* in short (a.k.a. of bounded geometry), i.e.,

$$N_X(R) := \sup\{|\operatorname{Ball}(x,R)| : x \in X\} < \infty$$

for every $R > 0$, where $\operatorname{Ball}(x,R) := \{y \in X : \operatorname{dist}(y,x) \leq R\}$. Associated with X are the *uniform Roe algebra* $\mathrm{C}^*_{\mathrm{u}}[X]$ and the *quasi-local algebra* $\mathrm{C}^*_{\mathrm{ql}}[X]$, prototypes of which are introduced in [Ro1]. These are C^*-subalgebras of the C^*-algebra $\mathbb{B}(\ell_2X)$ of bounded operators on the Hilbert space ℓ_2X. For $R > 0$, we denote by

$$\mathbb{C}^R_{\mathrm{u}}[X] := \{u \in \mathbb{B}(\ell_2X) : \langle u\delta_x, \delta_y\rangle = 0 \text{ whenever } \operatorname{dist}(x,y) > R\}$$

the set of all operators with *propagation at most* R. The uniform Roe algebra $\mathrm{C}^*_{\mathrm{u}}[X]$ is the norm closure of the $*$-algebra $\bigcup_{R>0}\mathbb{C}^R_{\mathrm{u}}[X]$ of finite propagation operators on ℓ_2X. An operator u on ℓ_2X is said to have ε*-propagation at most* R if it satisfies $\|1_A u 1_B\| \leq \varepsilon$ whenever $A, B \subset X$ are such that $\operatorname{dist}(A,B) > R$. Here, $1_A \in \mathbb{B}(\ell_2X)$ stands for the orthogonal projection from ℓ_2X onto ℓ_2A for any $A \subset X$. An operator $u \in \mathbb{B}(\ell_2X)$ is *quasi-local* if it has finite ε-propagation for all $\varepsilon > 0$. The quasi-local algebra $\mathrm{C}^*_{\mathrm{ql}}[X]$ is the C^*-algebra consisting of all quasi-local operators. It is plain to see that $\mathrm{C}^*_{\mathrm{u}}[X] \subset \mathrm{C}^*_{\mathrm{ql}}[X]$. The uniform Roe algebras and the quasi-local algebras have different advantages. Generally speaking, an operator in $\mathrm{C}^*_{\mathrm{u}}[X]$ is easier to handle than that in $\mathrm{C}^*_{\mathrm{ql}}[X]$, but it is harder to tell if a given operator u belongs to $\mathrm{C}^*_{\mathrm{u}}[X]$. Thus the problem whether they coincide or not has caught considerable attention (p. 20 in [Ro2],

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